# Lab 9 - Frequency Response Masking Filters (Optional)

For given passband ripple and stopband attenuation, the filter order, as well as the number of non-trivial coefficients, of a direct form FIR filter, is inversely proportional to the transition width. Therefore, sharp filters suffer from high computational complexity due to the high filter orders. Lim [1] proposed a frequency response masking (FRM) FIR filter structure in 1986 for the design of sharp FIR filters with low computational complexity. The structure was thereafter modified and improved by many researchers to achieve even further computation savings, and has been extended to the design of various types of filters, such as half-band filters, multirate filters, recursive filters and two-dimensional filters.

## 9.1 FRM Structure

Just as the name implies, FRM uses masking filters to obtain the desired frequency responses. The overall structure of an FRM is a subfilter network, consisting of three subfilters, as shown in Figure 9.1. The three subfilters are a prototype band edge shaping filter (or simply called prototype filter) with z-transform transfer function $F(z)$ and two masking filters with $z$-transform transfer functions $G_1(z)$ and $G_2(z)$, respectively; their corresponding frequency responses are denoted as $F(e^{j\omega})$, $G_1(e^{j\omega})$, and $G_2(e^{j\omega})$, respectively. A fourth filter function is realized by using a delay-complementary counterpart.

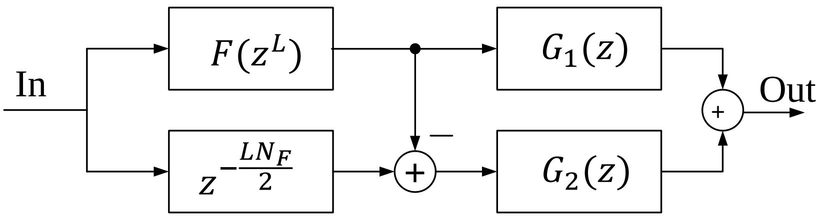


Figure 9.1: Structure of the frequency response masking filter.

In FRM structure, every delay of the prototype band edge shaping filter $F(z)$ is replaced by $L$ delays. The frequency response of the resulting filter, $F(z^L)$, is a frequency compressed (by a factor of $L$) and periodic version of $F(z)$, as shown in Figures 9.2 (a) and (b). For this reason, the number of delays replacing one delay in the impulse response of the band edge shaping prototype filter, $L$, is referred to as the compression factor of the FRM filter.

As a result of the delay element replacement, the transition band of $F(z)$ is mapped to $L$ transition bands with transition width shrunk by a factor of $L$. The complementary filter of $F(z^L)$ is obtained by subtracting $F(z^L)$ from a pure delay term, $z^{-LN_F/2}$, where $N_F$ is the order of $F(z)$. To avoid a half delay, $N_F$ is chosen to be even. Thus, the entire frequency region from DC to Nyquist frequency is decomposed into $L$ bands; $F(z^L)$ and its complement hold alternate band respectively, as shown in Figures 9.2 (b).

Two masking filters $G_1(z)$ and $G_2(z)$ are used to filter (i.e., mask) the outputs of $F(z^L)$ and its complement. The desired frequency bands of $F(z^L)$ and its complement are kept and then combined to obtain the final frequency response. The overall $z$-transform transfer function of the FRM structure, $H(z)$, is thus given by,

$$H(z) = F(z^L)G_1(z) + \left[z^{-\frac{LN_F}{2}} - F(z^L)\right]G_2(z) \tag{9.1}$$

In a synthesis problem, the passband and stopband edges of the overall filter $H(z)$, denoted $\omega_p$ and $\omega_s$, respectively, are given, whereas the passband and stopband edges of the prototype filter $F(z)$, denoted $\omega_p^F$ and $\omega_s^F$, have to be determined. From Figure 9.2 b, it can be seen that the transition band of the overall filter $H(z)$ may be generated either from $F(z^L)$ or from its complement.

We denote the two cases as Case A and Case B. The filter $F(z^L)$ (or its complement) is thus called periodic band edge shaping filter, or simply periodic filter or band edge shaping filter. We use lowpass filters as examples to illustrate the FRM technique. The extension of the technique to other frequency selective filters is straightforward.

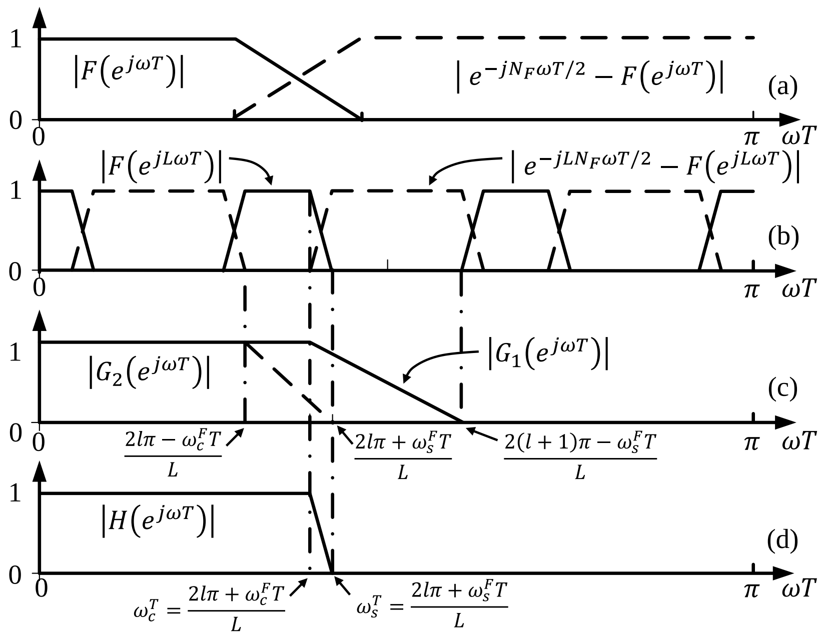


Figure 9.2: Frequency response of FRM subfilters of Case A.

## 9.1.1 Case A

In case A, the transition band of $H(z)$ is formed by the $l$-th transition band of $F(z^L)$, as shown in Figure 9.2, i.e.,

$$\omega_p = \frac{2l\pi + \omega_p^F}{L} \quad \text{and} \quad \omega_s = \frac{2l\pi + \omega_s^F}{L} \tag{9.2}$$

respectively. For a lowpass filter $H(z)$, the band edge of the prototype filter $F(z)$ must have $0 < \omega_p^F < \omega_s^F < \pi$, and meanwhile, we have $\omega_p < \omega_s$ and $l$ must be an integer. According to (9.2), there is only one value of $l$ that can meet these requirements if there is a solution, i.e., we must select $l = \lfloor L\omega_p/2\pi \rfloor$, where $\lfloor x \rfloor$ is the largest integer less than or equal to $x$.

Thus, for Case A, we have

$$\omega_p^F = L\omega_p - 2l\pi, \quad \omega_s^F = L\omega_s - 2l\pi \quad \text{for} \quad l = \lfloor L\omega_p/2\pi \rfloor \tag{9.3}$$

## 9.1.2 Case B

In Case B, the transition band of $H(z)$ is formed by the $(l-1)$-th transition band of $F(z^L)$'s complement, as shown in Figure 9.3, i.e.,

$$\omega_p = \frac{2l\pi - \omega_s^F}{L} \quad \text{and} \quad \omega_s = \frac{2l\pi - \omega_p^F}{L} \tag{9.4}$$

respectively. For the similar requirements as in Case A, the only value of $l$ if there is a solution is $l = \lceil L\omega_s/2\pi \rceil$ , where $\lceil x \rceil$ is the smallest integer larger than or equal to $x$. Thus, for Case B, we have

$$\omega_p^F = 2l\pi - L\omega_s, \quad \omega_s^F = 2l\pi - L\omega_p \quad \text{for} \quad l = \lfloor L\omega_p/2\pi \rfloor \tag{9.5}$$

Once the band edges of the prototype filter $F(z)$ are determined, for a given $L$, the band edges of the masking filters $G_1(z)$ and $G_2(z)$ may be chosen to keep the desired bands to obtain the overall responses. The band edge values of the masking filters are shown in Figures 9.2 (c) and 9.3 (c), respectively, for Case A and Case B.

Overall, Figure 9.2 shows a Case A example, whereas Figure 9.3 shows a Case B example.

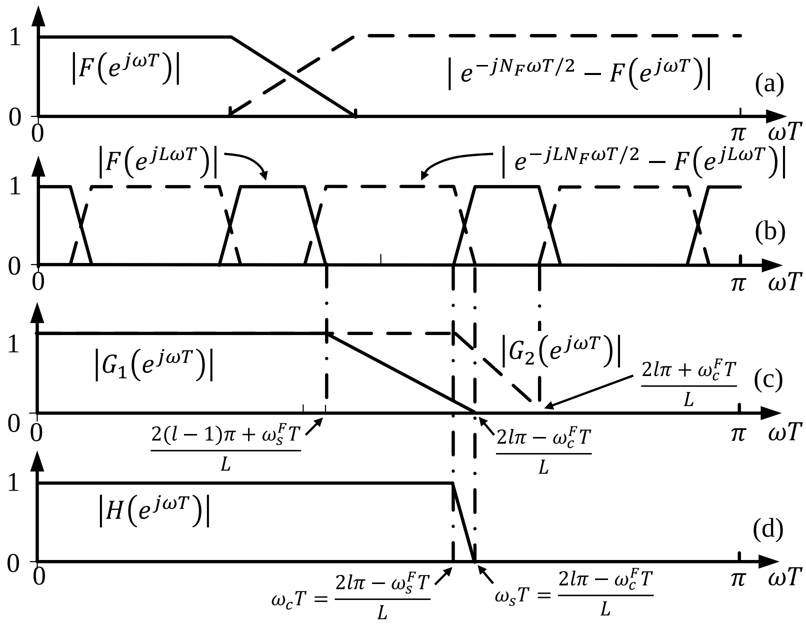

Figure 9.3: Frequency response of FRM subfilters of Case B.

## 9.2 Conditions for Feasible FRM Solutions

The conditions derived for Case A and Case B in sections 9.1.1 and 9.1.2 assume that a solution is available for a given set of $\omega_p, \omega_s$ and $L$. It is possible that neither Case A nor Case B generates a feasible solution for a combination of $\omega_p, \omega_s$ and $L$.

First, obviously only when $L > 2\pi/\omega_p$ and $L > 2\pi/(\pi - \omega_s)$, may FRM technique be considered; otherwise, the passband of the filter is too narrow or too wide to be obtained by combining bands from both band edge shaping filter and its complement. Besides the above condition, from (9.2), (9.4), and the fact that $0 < \omega_p^F < \omega_s^F < \pi$, and $\omega_p < \omega_s$, for a lowpass design, we can further find that $L$ must satisfy

$$\left\lceil \frac{L\omega_p}{\pi} \right\rceil \neq \left\lfloor \frac{L\omega_s}{\pi} \right\rfloor, \tag{9.6}$$

so that either Case A or Case B, but not both, leads to a feasible solution. When (9.6) is violated, no feasible solution exists.

## 9.3 Selection of $L$

Since the transition width of the prototype filter $F(z)$ is $L(\omega_s - \omega_p)$ for given $\omega_p$ and $\omega_s$, and the sum of the transition width of $G_1(z)$ and $G_2(z)$ is $2\pi/L$, the complexity of $F(z)$ decreases with increasing $L$, while the complexity of $G_1(z)$ and $G_2(z)$ increases with increasing $L$. It has shown that the minimum arithmetic complexity in terms of the number of multipliers is achieved by selecting

$$L = \frac{1}{\sqrt{2\beta(\omega_s - \omega_p)/\pi}} \tag{9.7}$$

where $\beta = 1$ if the subfilters are optimized separately, and $\beta = 0.6$ if the subfilters are optimized jointly.

## 9.4 Computational Complexity

The basic FRM structure utilizing the complementary periodic filter pairs and masking filters synthesizes sharp filter with low arithmetic complexity. When

$$\omega_s - \omega_p < 0.126\pi \text{ rad},$$

the arithmetic complexity saving is achieved by using FRM structure, compared with the direct form FIR filters. In a typical design with $\omega_s - \omega_p < 0.002$ rad, the FRM structure reduces the number of multipliers by a factor of 10, and the price paid for the arithmetic saving is a less than 5% increase in the overall filter order.

# 9.5 Design Procedure

For a given set of lowpass filter specifications, including passband edge, stopband edge, passband ripple and stopband ripple $\omega_p, \omega_s, \delta_p$ and $\delta_s$, respectively, a general design procedure, where all the subfilters are optimized separately, is given as follows:

**Step 1**. Determine the optimum $L$ that minimizes the filter complexity according to (9.7). If the selected $L$ violates (9.6), increase or decrease $L$ by 1.

**Step 2**. Determine the passband and stopband edges of the prototype band edge shaping filter according to (9.3) or (9.5).

**Step 3**. Determine the passband and stopband edges of the masking filters according to the parameters given in Figure 9.2 (c) or Figure 9.3 (c).

**Step 4**. Design the prototype band edge shaping filter and masking filters with 50% ripple margins.

**Step 5**. Verify the frequency response of the resulting FRM filter. If the specification has been satisfied, then the design is successful; otherwise, increase the ripple margin, and go to **Step 4**.

# 9.6 Design Exercise

Use FRM technique to design an FIR lowpass filter with passband edge, stopband edge, passband ripple and stopband ripple 0.4 rad, 0.402 rad, 0.01 and 0.001, respectively.

INLAB REPORT: Show the parameters of the design, for example, $L$, bandedges of the subfilters, filter lengths of the subfilters, etc. State if the design is a CASE A or CASE B design. Submit the plot of the frequency responses of the subfilters and the overall FRM filter. Comment on the complexity of the designed filter, compared with the direct form design.